

Vaani Benchmark V1.0: An Inclusive Multimodal Benchmark Dataset for Hindi

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Abstract

Benchmarking is critical for the systematic evaluation and comparison of automatic speech recognition (ASR) systems. While several open-source datasets are available for Hindi ASR, existing benchmarks remain limited in geographic diversity, demographic representation, and transcription robustness. We introduce an inclusive, multimodal Hindi ASR benchmark collected from 104 districts across India. The dataset consists of spontaneous speech elicited using image prompts and recorded in real-world acoustic conditions across diverse demographic groups. Each audio segment is annotated with three independent transcriptions, enabling multi-reference evaluation that accounts for permissible orthographic and lexical variations. This design supports more robust, inclusive, and realistic ASR evaluation. We benchmark multiple open-source and proprietary ASR models and report their comparative performance on the benchmark dataset.

Index Terms: speech recognition, dataset, benchmarks

1. Introduction

Benchmarking is a critical stage in the model development process, as it evaluates model performance under unseen conditions using standardized benchmark datasets. This process reveals the strengths and limitations of models, enables meaningful comparison across systems, and informs decisions about their real-world applicability. State-of-the-art (SotA) Automatic Speech Recognition (ASR) systems continue to struggle with significant speech variability arising from factors such as gender, age, speech impairments, race, and accent differences [1, 2]. It is therefore essential that benchmark datasets capture the full range of variations a model is likely to encounter. In a linguistically diverse country such as India, languages vary significantly across regions, and even within the same language there are substantial geographic and dialectal differences [3]. Careful consideration is thus required to ensure that benchmark datasets provide adequate and diverse representation of these variations.

Several benchmark datasets have been released to evaluate Hindi ASR either as part of larger corpora or through independent initiatives, such as IndicSUPERB [4], LAHAJA [5] and Vistaar [6]. LAHAJA comprises both read and extemporaneous speech across a diverse range of topics and use cases, totaling 12.5 hours of Hindi audio collected from 132 speakers spanning 83 districts of India. In addition, multiple data collection efforts—including FLEURS [7], CommonVoice [8], MUCS [9], GramVaani [10], and RESPIN [11]—have released evaluation sets that can be used to benchmark ASR performance. Despite the availability of numerous datasets, significant gaps remain in

open benchmark datasets, particularly in terms of comprehensive regional and dialectal coverage.

According to the 2011 Census of India, Hindi is the most widely spoken language in the country, with 43.63% of the population reporting Hindi or its dialects as their first language [3]. Hindi and its dialects are spoken across numerous states and Union Territories, forming a linguistic continuum rather than a single homogeneous language. Consequently, evaluating an ASR system designed for such a broad geographic and linguistic landscape requires benchmark datasets that adequately capture this regional and dialectal diversity. A benchmark dataset with a limited number of speakers may fail to capture demographic diversity adequately, resulting in incomplete or unfair evaluation of ASR systems. However, many existing benchmarks lack sufficient coverage of these diverse speech varieties.

The widely used ASR evaluation metric, Word Error Rate (WER), is highly sensitive to minor variations in reference transcriptions [12]. For the same audio segment, multiple valid transcriptions may exist [13]. Evaluating an ASR system against a single reference transcription can therefore unfairly penalize the model if this inherent subjectivity is not accounted for. Several approaches have been proposed for measuring WER in a multi-reference setup, and these studies show significant variation in measured error rates when using multiple valid transcripts versus a single reference transcript [12, 13]. However, most Hindi benchmark datasets provide only one reference transcription per audio segment and do not capture this transcriptional variability. Alternative synthetic approaches that rely on large language models to generate variations may lack cultural and regional grounding, and can introduce hallucinated content that undermines data reliability [14, 15].

Code-switching is prevalent in Indic languages [16]. Many ASR systems handle code-switched speech in different ways: some output all words in the primary script, while others preserve words in their native scripts [17]. However, several issues exist in current benchmarks. First, they often do not adequately capture code-switched speech. Second, in cases where code-switching is present, benchmarks may provide transcriptions either entirely in the primary script or in mixed/native scripts. This inconsistency creates challenges for fair and reliable evaluation of ASR systems in code-switching scenarios.

Model performance is highly susceptible to noise, and recognition accuracy can vary significantly with different types and levels of background noise [18]. Evaluating ASR systems under noisy conditions is therefore essential for understanding their suitability for real-world applications. Noise event tags present in benchmarking datasets can provide valuable insights into the conditions under which a model is evaluated. However, many existing benchmarks lack explicit annotations of noise

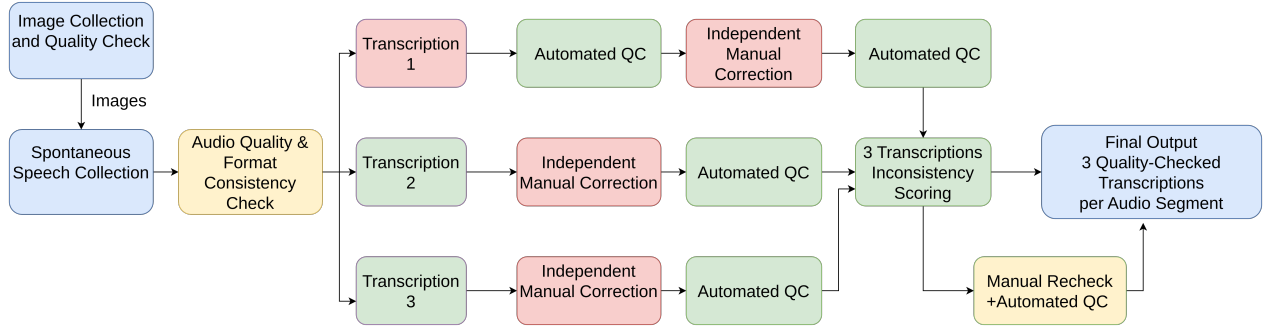


Figure 1: *Vaani Benchmark Data Preparation Process*

events, making it difficult to systematically analyze model robustness across diverse and realistic acoustic scenarios.

Recent models are increasingly moving toward multimodal architectures [19, 20]. Evaluating such models in multimodal scenarios requires datasets that contain aligned multimodal data. However, there is a lack of benchmark datasets for Indic languages that jointly combine modalities such as audio, image, and text. This gap limits systematic evaluation of multimodal models in Indic-language settings.

To address these challenges, we introduce a benchmark dataset for evaluating Hindi ASR systems. We present a multimodal benchmark comprising image and speech data, with three independent reference transcriptions for each audio segment. The dataset is collected from 104 districts across 22 states and Union Territories of India, capturing substantial geographic and linguistic diversity.

2. Vaani Benchmark

The benchmark targets systematic assessment of Hindi ASR robustness by pairing spoken utterances with multiple human references. It comprises 20.64 hours of spontaneous speech collected from an inclusive, geographically distributed speaker population spanning 104 districts across 22 Indian states and Union Territories, enabling fine-grained analysis of regional, acoustic, and socio-linguistic variations.

2.1. Collection and Preparation process

The data were collected as part of the Vaani Project [21] and comprise spontaneous speech elicited by asking speakers to describe images. The image set includes both district-specific images and images shared across multiple districts. Carefully designed guidelines were followed during image collection to ensure that local contexts were accurately captured. All images were newly captured specifically for this initiative. Image collection was completed prior to the speech data collection, and all images underwent rigorous quality control procedures, including automated and manual checks, to ensure high quality and the absence of any personally identifiable information (PII).

During the data collection process, speakers were shown up to 50 images and asked to verbally describe whatever came to mind upon viewing each image in the language spoken at home. Data collection was conducted through a mobile application. The collected data underwent a multi-stage quality control process. Initially, audio recordings were evaluated for quality and format consistency. Data collection was carried out with the support of vendors who coordinated the process at the district

level and were responsible for the first level of quality control. Each speaker contributed up to 15 minutes of accepted speech data.

As part of the Vaani Project, approximately 200 hours of spontaneous speech data were collected across 165 districts from 31 states and union territories after completing the quality control process. From this corpus, a benchmark dataset was constructed. During benchmark construction, sampling was performed within each district, with a maximum of 15 minutes of accepted speech selected per district. Within each district, utterances were sampled at the speaker level, with a maximum of five utterances per speaker. If a speaker had fewer than five utterances, all available utterances were included. For speakers with five or more utterances, up to five distinct images associated with the speaker were randomly selected; if fewer than five unique images were available, all were used. One utterance was randomly sampled from each selected image. If this process produced fewer than five utterances, additional utterances were randomly sampled from the same speaker—excluding previously selected ones—until five utterances were obtained. Finally, the selected segments were removed from the main dataset to prevent overlap between the benchmark and the remaining data.

As shown in Figure 1, the audio segments are initially transcribed by a single human transcriber and two different ASR systems. The manually generated transcription then undergoes consistency checks and automated quality validation. Subsequently, each transcription for an audio segment is independently reviewed by separate transcribers for correction, quality assurance, and verification of audio events. Well-defined transcription guidelines are strictly followed throughout the process, and transcribers for each district are selected from the same district to ensure familiarity with local dialects. Code-switched content is transcribed in both the native script and the original script, and non-speech audio events and noise instances are explicitly annotated alongside the spoken content.

Algorithm 1 Approach 3: Multi-reference Alignment-based WER Computation

Require: Dataset $\mathcal{D}$ with three reference transcripts $\{R_1, R_2, R_3\}$ and hypothesis transcript H for each utterance

Ensure: Multi-reference Word Error Rate (WER)

```

1: Initialize  $S \leftarrow 0, I \leftarrow 0, D \leftarrow 0, N \leftarrow 0$ 
2: for each utterance  $u \in \mathcal{D}$  do
3:   worderrors  $\leftarrow$  empty dictionary
4:   deleted_per_ref  $\leftarrow$  empty list
5:   Initialize  $I_u \leftarrow 0$ 
6:   for each reference transcript  $R_i \in \{R_1, R_2, R_3\}$  do
7:     Compute word-level alignment between  $R_i$  and  $H$ 
8:     Extract aligned reference words ref_words and hypothesis words hyp_words
9:     deleted_words  $\leftarrow \emptyset$ 
10:    for each alignment chunk do
11:      if chunk type is equal then
12:        Mark corresponding hypothesis words as correct
13:      else if chunk type is insert then
14:        Mark corresponding hypothesis words as insert
15:      else if chunk type is substitute then
16:        Mark corresponding hypothesis words as substitute
17:      else if chunk type is delete then
18:        Add deleted reference words to deleted_words
19:      end if
20:    end for
21:    Append deleted_words to deleted_per_ref
22:  end for
23:  Final word-level decision:
24:  for each hypothesis word  $w$  do
25:    if  $w$  is marked correct in any reference then
26:      Ignore  $w$  (not an error)
27:    else if  $w$  is marked substitute then
28:       $S \leftarrow S + 1$ 
29:    else if  $w$  is marked insert then
30:       $I_u \leftarrow I_u + 1$ 
31:    end if
32:  end for
33:  Compute true deletions as intersection of deletions across all references:

```

$$D_u = \bigcap_{i=1}^3 \text{deleted_per_ref}[i]$$

```

34:   $D \leftarrow D + |D_u|$ 

```

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35:  Compute effective reference length:

```

$$N_u = |H| + |D_u| - I_u$$

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36:   $N \leftarrow N + N_u$ 

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37:   $I \leftarrow I + I_u$ 

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38: end for

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39: Return final WER:

```

$$\text{WER} = \frac{S + I + D}{N}$$

After the first level of correction, transcription supervisors randomly audit approximately 10% of the transcriptions. If issues are identified, the affected transcriptions are returned to the same transcribers for correction. In cases where consistent errors are observed in a particular transcriber's work, the entire set of data assigned to that transcriber is reassigned to another transcriber for correction. Following another round of automated checks, consistency across the three transcriptions is evaluated. If significant discrepancies are observed among the three transcriptions for an audio segment, the segment is sent for manual validation to three independent transcribers.

2.2. The datasets

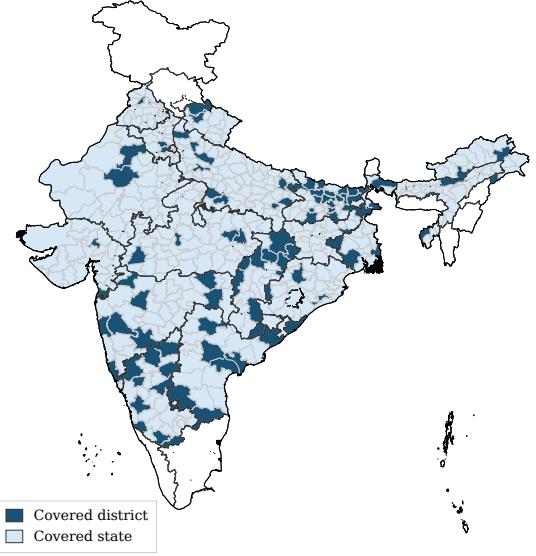


Figure 2: Districts and States represented in the Vaani Benchmark.

Following rigorous quality checks, the benchmark dataset includes speech data from 3,252 speakers, amounting to 20.64 hours of audio across 104 districts in 22 states and Union Territories. The dataset was collected using 8,315 images, with each audio segment transcribed by three independent transcribers. To support open research while preserving the integrity of the ASR leaderboard, 50% of the data from each district is released publicly, while the remaining data is retained as a closed evaluation set. The benchmark dataset is metadata-rich, providing unique speaker identifiers along with detailed attributes such as the languages known to each speaker, the PIN code of their native district, state information, and gender.

The dataset enables multi-reference evaluation of Hindi ASR systems, including performance analysis under code-switching scenarios and diverse noise conditions. It further supports benchmarking of audio-based and text-based image retrieval tasks, as well as evaluation of speaker identification and district identification systems. Given the substantial variation in Hindi dialects and accents across states and districts, the included geographical tags facilitate systematic analysis of geographical bias in ASR systems.

It has been observed that even after multiple rounds of quality control, a residual WER of 10–15% persists across transcription sets. Given that the data underwent several iterations of

Table 1: WER Scores for Different ASR Models

S.No	Model	Type	Approach 1	Approach 2	Approach 3	District Mean $\pm$ Std
1	Vaani Fast Conformer	Open	17.5	14.0	10.6	15.2 $\pm$ 4.1
2	Gemini-3.1-Pro	Closed	18.8	15.1	11.9	16.3 $\pm$ 4.3
3	Sarvam Saaras v3	Closed	20.3	16.9	13.7	18.3 $\pm$ 4.6
4	Indic-conformer-600m-multilingual	Open	21.0	17.5	14.2	18.9 $\pm$ 5.2
5	Sarvam Saarika v2.5	Closed	21.1	17.5	14.4	18.7 $\pm$ 4.7
6	Google Chirp 3	Closed	21.3	17.9	14.4	18.3 $\pm$ 12.7
7	Gemini-3-Flash-Preview	Closed	21.8	18.2	14.9	19.7 $\pm$ 5.2
8	Gemini-3.5-Flash	Closed	22.6	19.0	15.7	19.9 $\pm$ 5.7
9	Gemini-3.1-Flash-Live	Closed	24.7	21.4	18.2	22.3 $\pm$ 6.1
10	Shrutam-HindiASR-1.0	Open	24.7	21.4	18.2	23.0 $\pm$ 5.7
11	Gemma-3n-E2B-IT	Open	26.2	22.9	19.7	24.0 $\pm$ 7.2
12	Azure Speech	Closed	26.5	23.2	18.6	25.2 $\pm$ 6.9
13	Gemma4-E2B-IT	Open	27.0	23.8	20.1	24.7 $\pm$ 6.0
14	Voxtral-Mini-3B-2507	Open	28.1	25.0	21.6	26.1 $\pm$ 7.4
15	Pingala-v1-universal	Open	29.1	25.6	22.5	26.6 $\pm$ 5.6
16	Data2vec_aqc	Open	29.7	26.4	23.1	28.1 $\pm$ 6.7
17	Gemma4-12b	Open	33.1	29.9	26.4	30.9 $\pm$ 12.4
18	OmniASR_LLM_1B	Open	33.1	29.9	26.4	31.0 $\pm$ 6.3
19	Whisper-large-v3	Open	33.5	30.3	27.1	32.0 $\pm$ 8.9
20	Vakyansh-wav2vec2	Open	43.8	41.5	38.7	43.4 $\pm$ 8.0
21	GPT-4o-Transcribe	Closed	70.9	69.5	65.3	69.6 $\pm$ 11.0

	Set 1	Set 2	Set 3
Set 1	–	10.51%	13.62%
Set 2	10.51%	–	12.91%
Set 3	13.62%	12.91%	–

Table 2: Pairwise inter-set agreement measured using Word Error Rate (WER)

both human and automated validation, this discrepancy is likely attributable to natural variability in how different transcribers perceive and render spoken content. Such variation reflects the inherent linguistic diversity of languages like Hindi and highlights a key limitation of single-reference WER-based evaluation, where multiple valid transcriptions may exist for the same utterance.

3. Evaluations

We evaluated multiple models on the proposed benchmark dataset, including both open-source and commercial systems. The evaluated models comprise base versions as well as fine-tuned variants, and include both monolingual and multilingual architectures. While the Vaani dataset inherently supports multimodal benchmarking—such as speech-image retrieval and visual grounding—due to its paired audio, image, and text architecture, in this work, we limit our evaluation exclusively to the foundational task of ASR. For this ASR evaluation, we report the WER, computed using three different evaluation protocols.

We evaluated a diverse set of automatic ASR models, including both open-source and commercial systems. The evaluated models include Indic-focused encoder-decoder systems such as `indic-conformer-600m-multilingual`[22]; self-supervised and representation-learning-based models such as `SPRING-INX_Hindi`[23] and `vakyansh-hindi-large-wav2vec2`[24]; multilingual and large-scale ASR mod-

els including `whisper-large-v3`[25], `omniASR_LLM_1B`[26], `Voxtral-Mini-3B-2507`[27], and `pingala-v1-universal`[28]; as well as emerging multimodal and LLM-based systems such as `Gemini3-flash`[29]. In addition, we evaluated commercial cloud-based ASR systems including `Google Chirp`[30], `Azure Speech`[31], `Sarvam saarika-v2.5`[32] and `GPT-4o Transcribe`[33] and `Sarvam saaras-v3`[34].

We use three approaches for the evaluation. In **Approach 1**, WER is computed independently for each of the three transcription sets (Set-1, Set-2, and Set-3). Cumulative WER of each set computed and the performance metric of this approach is reported as the mean WER across the three sets. In **Approach 2**, for each segment, errors are calculated separately with respect to all three transcriptions, and the minimum error count among the three is selected. These minimum segment-level errors are then aggregated to compute the overall WER. In **Approach 3** (Algorithm 1), errors are computed by accounting for alternative word-level realizations across the three transcriptions. This approach explicitly models variability arising from different ways a word may be represented in Hindi depending on the listener’s perception.

The Vaani benchmark dataset includes district-level metadata indicating the geographic location from which the data was collected. This information enables the analysis of potential performance biases associated with geography. To facilitate geography-based performance evaluation, we computed model performance separately for each district and then calculated the mean and variance of performance across districts.

After evaluation, the Word Error Rate (WER) obtained using Approaches 1, 2, and 3, along with the mean and standard deviation across districts computed using Approach 2, are presented in Table 1. The results show considerable variation in WER across the three approaches, indicating the presence of multiple valid representations of the same audio segments. In the performance basis evaluation, we see performance difference across the models.

4. Conclusion

We present an inclusive, multimodal Hindi ASR benchmark comprising speech data collected from 104 districts, with three independent reference transcriptions for each segment. We evaluate multiple open-source and proprietary ASR models on this dataset. Our analysis reveals substantial differences in Word Error Rate (WER) when using multi-reference evaluation compared to the traditional single-reference setting, highlighting the impact of permissible orthographic variations on performance metrics. Diversity analysis further shows performance disparities across districts, with standard deviation indicating regional biases in some models. In future work, we plan to extend the benchmark to additional languages, expand its geographic coverage, and evaluate tasks such as multimodal ASR, image retrieval, and model robustness under different noise conditions using noise tags.

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